

TYRKA, Russian Federation

WMO: 305260

Lat: 54.508N

Lon: 107.138E

Elev: 2135

StdP: 13.60

Time Zone: 8.00 (E08)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.9	-41.2	-49.2	0.3	-45.7	-44.2	0.5	-40.2	15.4	-10.2	9.7	2.9	0.3	310	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	82.8	62.8	79.2	61.4	76.0	60.3	65.7	76.0	64.0	74.5	62.2	72.2	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	91.4	69.3	60.3	84.6	67.5	58.3	78.7	65.9	31.8	75.4	30.4	74.5	29.1	72.3	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	10.1	8.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.4	-15.1	-5.3	10.9	30.1	43.2	56.1	62.1	57.6	42.7	26.4	3.3	-8.7
	DBStd	28.44	13.83	12.50	12.25	8.69	7.47	7.10	4.72	5.58	7.37	9.49	14.25	14.58
	HDD50	9836	2019	1548	1212	596	237	24	0	10	240	731	1400	1819
	HDD65	14476	2484	1968	1677	1046	677	276	113	235	670	1196	1850	2284
	CDD50	875	0	0	0	0	25	207	377	246	20	0	0	0
	CDD65	39	0	0	0	0	0	9	25	5	0	0	0	0
	CDH74	1085	0	0	0	0	27	378	508	164	8	0	0	0
	CDH80	219	0	0	0	0	3	86	105	24	1	0	0	0
	WSAvg	2.4	1.4	1.7	2.5	3.7	4.2	2.9	2.6	2.3	2.4	2.3	1.7	1.5
	PrecAvg	14.0	0.5	0.3	0.3	0.6	1.1	2.0	2.9	2.9	1.5	0.6	0.7	0.6
(15) Precipitation	PrecMax	17.2	1.9	0.6	0.9	1.2	2.0	4.3	5.8	4.8	3.3	1.5	1.2	1.8
	PrecMin	11.0	0.1	0.0	0.0	0.2	0.3	0.8	1.5	1.3	0.6	0.1	0.3	0.2
	PrecStd	1.7	0.3	0.1	0.2	0.2	0.4	0.8	1.0	1.1	0.7	0.3	0.2	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	31.5	48.7	64.1	79.3	87.4	87.2	84.3	74.2	58.2	37.5	25.9
		MCWB	19.1	26.2	36.7	46.4	57.2	62.8	64.1	63.9	56.9	45.0	32.5	23.9
	2%	DB	15.0	25.6	41.4	56.7	71.0	81.7	83.3	78.6	66.4	50.2	32.1	20.7
		MCWB	13.7	22.1	32.6	42.7	51.7	61.1	63.4	61.4	53.4	40.3	29.2	19.1
	5%	DB	10.7	21.1	36.1	51.3	66.0	78.2	79.8	75.3	60.7	44.3	27.9	17.3
		MCWB	9.8	18.5	29.7	39.8	49.0	59.3	63.1	60.4	50.9	37.4	25.5	16.0
	10%	DB	6.3	15.6	31.6	46.2	61.3	74.4	76.3	71.4	55.9	39.9	23.8	13.0
		MCWB	5.5	13.6	26.6	36.5	46.9	58.5	62.0	59.7	48.3	34.9	21.9	12.0
	0.4%	WB	19.4	26.6	37.3	47.6	57.4	66.0	68.8	67.1	58.6	45.7	33.7	24.3
		MCDB	20.9	30.7	47.8	60.5	78.9	78.5	78.7	78.6	70.2	57.3	36.8	25.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	13.7	22.4	33.1	43.9	53.6	63.5	66.6	64.4	55.0	40.9	29.2	19.4
		MCDB	15.0	25.3	40.2	55.2	66.6	76.9	76.4	74.1	64.1	48.6	32.0	20.7
	5%	WB	9.9	18.3	29.9	40.5	50.5	61.2	64.9	62.4	52.0	38.0	25.5	16.1
		MCDB	10.8	21.3	35.9	50.2	64.1	74.8	75.0	71.1	59.0	43.6	27.9	17.3
	10%	WB	5.6	13.6	27.0	37.2	47.6	59.1	63.3	60.4	49.0	35.3	21.9	12.2
		MCDB	6.4	15.6	31.5	45.8	59.6	72.4	73.4	69.0	55.0	39.7	23.7	13.1
(35) Mean Daily Temperature Range	5% DB	MDBR	23.6	31.2	32.2	25.2	24.9	26.1	23.5	23.3	21.6	20.6	21.3	20.6
		MCDBR	22.5	33.9	35.3	33.8	36.0	35.6	32.1	32.0	31.3	29.5	19.8	17.0
		MCWBR	21.0	29.9	26.6	20.3	19.0	16.1	14.7	15.5	18.3	20.3	16.6	16.0
	5% WB	MCDBR	22.2	32.7	33.2	32.3	33.1	31.0	25.6	24.8	28.0	26.8	19.7	17.0
		MCWBR	21.0	28.4	25.3	19.9	18.4	15.4	13.6	13.8	17.5	19.2	16.6	16.1
(40) Clear-Sky Solar Irradiance	taub		0.263	0.293	0.316	0.354	0.399	0.440	0.433	0.382	0.335	0.307	0.281	0.259
	taud		2.066	2.052	2.078	2.204	2.211	2.168	2.229	2.361	2.449	2.413	2.171	2.032
	Ebn at Noon		210	245	269	275	268	257	256	265	265	246	202	180
	Edh at Noon		22	31	39	40	42	44	41	34	27	22	19	18
(44) All-Sky Solar Radiation	RadAvg		330	663	1158	1550	1662	1800	1660	1408	966	570	314	230
	RadStd		24	26	39	105	133	117	160	65	74	71	25	18

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	-3.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page